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Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech S1 (Special Improvement) Examination January 2021 (2019 scheme)

Course Code: MAT101

Course Name: LINEAR ALGEBRA AND CALCULUS

(2019-Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

- 1 Determine the rank of the matrix $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 3 & -1 \\ 5 & 3 & -2 \end{bmatrix}$ (3)
- 2 What kind of conic section is represented by the quadratic form $7x_1^2 + 6x_1x_2 + 7x_2^2 = 200$ Transform it into canonical form. (3)
- 3 Find the derivative of $w = x^2 + y^2$ with respect to t along the path $x = at^2, y = 2at$. (3)
- 4 Let $f(x, y) = \sqrt{3x + 2y}$, find the slope of the surface $z = f(x, y)$ in the y -direction at the point $(2, 5)$. (3)
- 5 Evaluate $\int_0^a \int_0^a \int_0^a (yz + xz + xy) dx dy dz$ (3)
- 6 Use polar co-ordinates to evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{\frac{3}{2}} dy dx$ (3)
- 7 Test the convergence of $\sum_{k=1}^{\infty} \frac{1}{(2k+3)^{17}}$ (3)
- 8 Examine whether the series convergence or not $\sum_{k=1}^{\infty} \frac{1}{(\ln(k+1))^k}$ (3)
- 9 Find the Maclaurin series of $\frac{1}{x+1}$ up to third degree term. (3)
- 10 Find the Fourier Half Range sine series of $f(x) = x$ in $0 < x < \pi$. (3)

PART B

Answer one full question from each module, each question carries 14 marks

Module-I

- 11 a) Test for consistency and solve the system of equations (7)

$$x + 2y - z = 3$$

$$3x - y + 2z = 1$$

$$2x - 2y + 3z = 2$$

$$x - y + z = -1$$

- b) Find the eigenvalues and eigenvectors of
- $\begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$
- (7)

- 12 a) For what values of a and b do the system of equations (7)

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + az = b$$

have i) no solution ii) unique solution iii) more than one solution.

- b) Find the matrix of transformation that diagonalize the matrix (7)

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}. \text{ Also write the diagonal matrix.}$$

Module-II

- 13 a) If
- $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$
- find the value of
- $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$
- . (7)

- b) If the local linear approximation of a function
- $f(x, y, z) = xy + z^2$
- at a point P is
- $L(x, y, z) = y + 2z - x$
- , find the point P. (7)

- 14 a) If
- $z = e^{xy}$
- ,
- $x = 2u + v$
- ,
- $y = \frac{u}{v}$
- find
- $\frac{\partial z}{\partial u}$
- . (7)

- b) Locate all relative extrema of
- $f(x, y) = 3x^2 - 2xy + y^2 - 8y$
- . (7)

Module-III

- 15 a) Evaluate
- $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$
- by reversing the order of integration. (7)

- b) Using triple integral find the volume of the solid in the first octant bounded by the coordinate planes and the plane
- $x + 2y + z = 6$
- . (7)

- 16 a) Find the mass and center of gravity of the triangular lamina with vertices $(0,0)$, $(0,1)$ and $(1,0)$ and density function $\delta(x,y)=xy$ (7)
- b) Evaluate $\iint_R x^2 dy dx$, where R is the region between $y=x$ and $y=x^2$ (7)

Module-IV

- 17 a) Discuss the convergence of the series (7)

$$(i) \sum_{k=1}^{\infty} \frac{k!}{k^k} \quad (ii) \sum_{k=1}^{\infty} \left(\frac{k}{k+1} \right)^{k^2}$$

- b) Examine the convergence and divergence of the series (7)

$$\frac{x}{1.3} + \frac{x^2}{3.5} + \frac{x^3}{5.7} + \dots$$

- 18 a) Test the convergence of $1 + \frac{1.3}{3!} + \frac{1.3.5}{5!} + \frac{1.3.5.7}{7!} + \dots$ (7)

- b) Prove that the series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{(k-2)}{k(k+1)}$ is conditionally convergent. (7)

Module-V

- 19 a) Obtain Fourier series for the function $f(x) = |\sin x|$ $-\pi < x < \pi$ (7)

- b) If $f(x) = \begin{cases} kx & ; 0 < x < \frac{\pi}{2} \\ k(\pi - x) & ; \frac{\pi}{2} < x < \pi \end{cases}$ then show that (7)

$$f(x) = \frac{4k}{\pi} \left(\frac{\sin x}{1^2} - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \dots \right).$$

- 20 a) Find the Fourier cosine series of $f(x)=x^2$ in $(0,\pi)$. Hence show that (7)

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

- b) Find the Fourier series for the function (7)

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 1-x & 1 < x < 2 \end{cases}$$

B-TECH S₁ (Special improvement) - Jan 2021

MAT101 (Linear algebra and Calculus)

PART A

①

No of non zero rows = Rank of a matrix .

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 3 & -1 \\ 5 & 3 & -2 \end{bmatrix}$$

using Gauss elimination

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 5R_1$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & -1 \\ 0 & 8 & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

no of non zero rows = 2 = Rank(A)

②

$$7x_1^2 + 6x_1x_2 + 7x_2^2 = 200$$

$$A = \begin{bmatrix} 7 & 3 \\ 3 & 7 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 7-\lambda & 3 \\ 3 & 7-\lambda \end{bmatrix}$$

$$|A - \lambda I| = 0 \Rightarrow (7-\lambda)^2 - 9 = 0$$

$$49 + \lambda^2 - 14\lambda - 9 = 0$$

$$= \lambda^2 - 14\lambda + 40 = 0$$

$$\lambda_1 = 4, \lambda_2 = 10$$

$$Q = \lambda_1 y_1^2 + \lambda_2 y_2^2$$

$$Q = 10y_1^2 + 4y_2^2 \quad \because (Q = 200)$$

$$200 = 10y_1^2 + 4y_2^2$$

$$100 = 5y_1^2 + 2y_2^2$$

$$\frac{y_1^2}{(20)^2} + \frac{y_2^2}{50} = 1, \text{ Ellipse}$$

For expressing x in terms of y we have to find eigen vectors.

For $\lambda = 10$

$$\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-3x_1 + 3x_2 = 0$$

$$x_1 = x_2$$

$$X = \begin{bmatrix} a \\ a \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} a$$

$$X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda = 4$

$$\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$3x_1 + 3x_2 = 0$$

$$x_1 = -x_2$$

$$X = \begin{bmatrix} -a \\ a \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} a$$

$$X_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\frac{x_1}{|x_1|} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \quad \frac{x_2}{|x_2|} = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$x = Xy$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \frac{y_1}{\sqrt{2}} - \frac{y_2}{\sqrt{2}} \\ \frac{y_1}{\sqrt{2}} + \frac{y_2}{\sqrt{2}} \end{pmatrix}$$

$$x_1 = \frac{y_1}{\sqrt{2}} - \frac{y_2}{\sqrt{2}}$$

$$x_2 = \frac{y_1}{\sqrt{2}} + \frac{y_2}{\sqrt{2}}$$

③

$$w = x^2 + y^2$$

$$\frac{dw}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$x = at^2$$

$$\frac{dx}{dt} = 2at$$

$$\frac{dy}{dt} = 2a$$

$$\begin{aligned} \frac{dw}{dt} &= 2at^2 \times 2at + 2 \times 2at + 2a \\ &= \underline{4a^2t^2 + 8a^2t} \end{aligned}$$

④

$$f(x, y) = \sqrt{3x+2y}$$

$$\frac{\partial z}{\partial y} = \frac{\partial (\sqrt{3x+2y})}{\partial y}$$

$$= \frac{1 \times 2}{2\sqrt{3x+2y}} = \frac{1}{\sqrt{3x+2y}}$$

at (2, 5)

$$= \frac{1}{\sqrt{16}} = \underline{\underline{\frac{1}{4}}}$$

⑤

$$\int_0^a \int_0^a \int_0^a (yz + xz + xy) dx dy dz$$

$$\int_0^a \int_0^a \left(yz x + \frac{x^2}{2} z + \frac{x^2}{2} y \right) dx dy dz = \int_0^a \int_0^a \left(ayz + \frac{a^2}{2} z + \frac{a^2}{2} y \right) dy dz$$

$$= \int_0^a \left(\frac{a^2 y^2}{2} + \frac{a^2}{2} zy + \frac{a^2}{2} \frac{y^2}{2} \right) dy dz =$$

$$= \int_0^a \left(\frac{a^3}{2} z + \frac{a^3}{2} z + \frac{a^4}{4} \right) dz$$

$$= \int_0^a \left(\frac{a^3}{2} \frac{z^2}{2} + \frac{a^3}{2} \frac{z^2}{2} + \frac{a^4}{4} z \right) dz$$

$$= \frac{a^5}{4} + \frac{a^5}{4} + \frac{a^5}{4} = \underline{\underline{\frac{3a^5}{4}}}$$

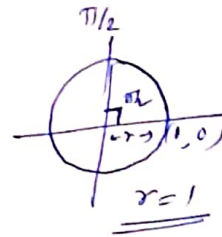
$$\textcircled{6} \int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)^{3/2} dy dx$$

$$x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2, \quad y/x = \tan \theta$$

$$\text{region } x = -1 \rightarrow 1$$

$$y = 0 \rightarrow \sqrt{1-x^2}$$

$$y^2 = (1-x^2) \Rightarrow x^2 + y^2 = 1$$



$$r = 0 \rightarrow 1, \quad \theta = 0 \rightarrow \pi/2$$

$$\int_{\theta=0}^{\pi/2} \int_{r=0}^1 (r^2)^{3/2} dr d\theta$$

$$= \int_{\theta=0}^{\pi/2} \int_{r=0}^1 r^3 dr d\theta$$

$$\int_{\theta=0}^{\pi/2} \left(\frac{r^4}{4} \right)_0^1 d\theta = \int_{\theta=0}^{\pi/2} \frac{1}{4} d\theta$$

$$\frac{1}{4} \left[\theta \right]_0^{\pi/2} = \frac{1}{4} \times \frac{\pi}{2}$$

$$= \frac{\pi}{8}$$

7 $\sum_{k=1}^{\infty} \frac{1}{(2k+3)^{17}}$

$$\sum a_k = \sum \frac{1}{(2k+3)^{17}}$$

$$s_k = \sum \frac{1}{k^{17}}$$

It is a p series with $p=17$

$p > 1 \therefore$ Convergent

$$1 = \lim_{k \rightarrow \infty} \frac{a_k}{a_k} = \lim_{k \rightarrow \infty} \frac{\frac{1}{(2k+3)^{17}}}{\frac{1}{2k^{17}}}$$

$$= \lim_{k \rightarrow \infty} \frac{\frac{1}{k^{17}} (2 + \frac{3}{k})^{17}}{\frac{1}{k^{17}}}$$

$$= \frac{1}{2^{17}} > 0$$

Finite and non-zero

8 $\sum_{k=1}^{\infty} \left[\frac{1}{\ln(k+1)} \right]^k$

Root test

$$u_k = \frac{1}{[\ln(k+1)]^k}$$

$$(u_k)^{1/k} = \left[\frac{1}{[\ln(k+1)]^k} \right]^{1/k} = \frac{1}{\ln(k+1)}$$

$$\text{Now } L = \lim_{k \rightarrow \infty} (u_k)^{1/k}$$

$$= \lim_{k \rightarrow \infty} \frac{1}{\ln(k+1)}$$

$$= \frac{1}{\infty} = 0 < 1$$

$\therefore$ converges by root test

9. $f(x) = \frac{1}{1+x}$

Maclaurian Series, $x=0$

$\therefore f(0) = 1$

$f'(x) = \frac{-1}{(1+x)^2}$ $f'(0) = -1$

$f''(x) = \frac{2}{(1+x)^3}$ $f''(0) = 2$

$f'''(x) = \frac{-6}{(1+x)^4}$ $f'''(0) = -6$

$$f(x) = \frac{1}{1+x} = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$= 1 - x + x^2 - x^3 + \dots$$

10. $f(x) = x$ $0 < x < \pi$

$L = \pi$

Half range sine series.

$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$

$= \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{\pi}$

$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$

$= \frac{2}{\pi} \int_0^{\pi} x \sin nx \, dx$

$= \frac{2}{\pi} \left[x \cdot \frac{\cos nx}{n} - \int 1 \cdot \frac{-\cos nx}{n} \right]_0^{\pi}$

$= \frac{2}{\pi} \left[x \cdot \frac{\cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^{\pi}$

$= \frac{2}{\pi} \left[-\frac{\pi \cos n\pi}{n} + \frac{\sin n\pi}{n^2} - \left(-\frac{x \cos 0}{n} + \frac{\sin 0}{n^2} \right) \right]$

$= \frac{2}{\pi} \left[-\frac{(-1)^n \pi}{n} + 0 - \left(-\frac{\pi}{n} + 0 \right) \right] = \frac{2}{\pi} \left[\frac{\pi}{n} - \frac{(-1)^n \pi}{n} \right]$

$\therefore f(x) = \sum_{n=1}^{\infty} \frac{2}{\pi} \left(\frac{\pi}{n} - \frac{(-1)^n \pi}{n} \right) \sin nx$

Module - I

11. a) $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 2 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix}$ $AX = B$

$$C = [A:B] = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \\ 2 & -2 & 3 & 2 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$= \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 2 & -2 & 3 & 2 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$= \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & 1 & 4 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - R_1$$

$$= \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & -3 & 2 & -4 \end{bmatrix}$$

$$R_4 \rightarrow 7R_4 - 3R_2$$

$$= \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -1 & -4 \end{bmatrix}$$

$$R_4 \rightarrow R_4 + R_3$$

$$= \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank of } A = \text{rank of } [A:B] = n.$$

Unique soln.

$$x + 2y - z = 3$$

$$-2y + 5z = -8$$

$$z = 4$$

$$x = -1$$

$$y = 4$$

$$z = 4$$

$$b) A = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 1 & 2 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & 3-\lambda \end{vmatrix} = 0.$$

$$-\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0.$$

$$\lambda = 3, 1, 2$$

$$\lambda = 3$$

$$(A - 3I) = \begin{bmatrix} -2 & 1 & 2 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$k = \frac{x_1}{-1} = \frac{-x_1}{0} = \frac{x_3}{-1}$$

$$\frac{x_1}{-1} = \frac{-x_2}{0} = \frac{x_3}{-1}$$

$$x = k \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix}$$

$$\lambda = 1$$

$$(A - I) = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \quad \frac{x_1}{1} = \frac{-x_1}{-1} = \frac{x_3}{-1}$$

$$k = \frac{x_1}{1} = \frac{-x_1}{-1} = \frac{x_3}{-1} \quad x_2 = k \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\lambda = 2$$

$$(A - 2I) = \begin{bmatrix} -1 & 1 & 2 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad \frac{x_1}{0} = \frac{-x_2}{-1} = \frac{x_3}{0}$$

$$k = \frac{x_1}{-1} = \frac{-x_2}{-1} = \frac{x_3}{-1} \quad x_3 = k \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$$

12. a) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & a \end{bmatrix} \quad B = \begin{bmatrix} 6 \\ 10 \\ b \end{bmatrix} \quad AX = B.$

$$C = [A : B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & a & b \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 0 & 0 & a-3 & b-10 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & a-3 & b-10 \end{bmatrix}$$

Before inequality, if

i) when $a=3$, $b \neq 10$, $[A:B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 6 & 1 & 2 & 4 \\ 0 & 0 & 0 & k \end{bmatrix}$ rank of $A \neq$ rank of $[A:B]$

System has no solution.

ii) when $a \neq 3$ and b any ~~any~~ value, $[A:B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 6 & 1 & 2 & 4 \\ 0 & 0 & k & m \end{bmatrix}$

rank of $A =$ rank of $[A:B] = n = 3$.

System has a unique solution.

iii) when $a=3$ and $b=10$, $[A:B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 6 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

rank of $A =$ rank of $[A:B] < n = 3$.

System has infinite solution.

b) $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ $|A - \lambda I| = 0 = \begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix}$

characteristic equation given as $-\lambda^3 + 18\lambda^2 - 45\lambda = 0$

$\lambda = 15, 0, 3$

For $\lambda = 15$

$[A - 15I] = \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix}$ $\frac{x_1}{-8-4} = \frac{-x_2}{-6-4} = \frac{x_3}{-6-8}$

$k = \frac{x_1}{80} = \frac{-x_2}{80} = \frac{x_3}{40}$ $x_2 = k \begin{bmatrix} 80 \\ -80 \\ 20 \end{bmatrix} = k \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

$$\lambda = 0$$

$$[A - 0I] = \begin{bmatrix} 0 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \quad \frac{x_1}{2} = \frac{-x_2}{-4} = \frac{x_3}{-2}$$

$$k = \frac{x_1}{5} = \frac{-x_2}{-10} = \frac{x_3}{10} \quad X = k \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\lambda = 3$$

$$[A - 3I] = \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \quad \frac{x_1}{4} = \frac{-x_2}{-4} = \frac{x_3}{2}$$

$$k = \frac{x_1}{-16} = \frac{-x_2}{8} = \frac{x_3}{16} \quad X = k \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$$

$$P = [X_1 \ X_2 \ X_3] = \begin{bmatrix} 2 & 1 & -2 \\ -2 & 2 & -1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$P^{-1}AP = D = \begin{bmatrix} 15 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Module II

13. a) $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$

Let $\frac{x}{y} = r$ $\frac{y}{z} = s$ $\frac{z}{x} = t$.

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial x}$$

$$= \frac{\partial u}{\partial r} \times \frac{1}{y} + \frac{\partial u}{\partial t} \times \frac{-z}{x^2}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial y}$$

$$= \frac{\partial u}{\partial r} \cdot \frac{-x}{y^2} + \frac{\partial u}{\partial s} \cdot \frac{1}{z}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial z}$$

$$= \frac{\partial u}{\partial s} \cdot \frac{-y}{z^2} + \frac{\partial u}{\partial t} \cdot \frac{1}{x}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \cdot \frac{x}{y} + \frac{\partial u}{\partial t} \cdot \frac{z}{x} + \frac{\partial u}{\partial r} \cdot \frac{x}{y} + \frac{\partial u}{\partial s} \cdot \frac{y}{z} - \frac{\partial u}{\partial s} \cdot \frac{y}{z} + \frac{\partial u}{\partial t} \cdot \frac{z}{x}$$

$$= 0$$

b) $L(x_0, y_0, z_0) = f(x_0, y_0, z_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + f_z(x_0, y_0)(z - z_0)$

$$f(x_0, y_0, z_0) = x_0 y_0 + z_0^2$$

$$f_x = y$$

$$f_y = x$$

$$f_z = 2z$$

$$L(x, y, z) = x_0 y_0 + z_0^2 + y_0(x - x_0) + x_0(y - y_0) + 2z_0(z - z_0)$$

$$= x_0 y_0 + z_0^2 + x y_0 - x_0 y_0 + x_0 y - x_0 y_0 + 2z_0 z - 2z_0^2$$

$$= -z_0^2 + 2z_0 z - x_0 y_0 + x_0 y + x y_0 \quad \text{--- (1)}$$

From Question $L(x, y, z) = y + 2z - x$.

Comparing coefficients, $-z_0^2 + -x_0 y_0 = 0$.

$$x_0 y_0 + z_0^2 = 0$$

$$2z_0 = 2$$

$$z_0 = 1$$

$$x_0 = 1 \quad y_0 = -1$$

P2

The point P is given by ordered triplet $(1, -1, 1)$

14. a) $z = e^{xy}$

$$x = 2u + v \quad y = \frac{u}{v}$$

~~xxxxxx~~ $u =$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$= ye^{xy} \cdot 2 + xe^{xy} \cdot \frac{1}{v}$$

$$= 2ye^{xy} + \frac{xe^{xy}}{v}$$

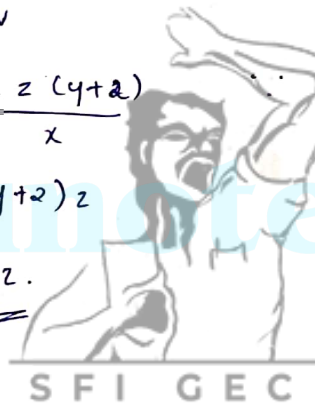
$$= 2yz + \frac{xz}{v}$$

$$= 2yz + \frac{xz(y+2)}{x}$$

$$= 2yz + (y+2)z$$

$$\frac{\partial z}{\partial u} = 3yz + 2z$$

$$=$$



$$u = vy$$

$$v = x - 2vy$$

$$x(v+2vy) = x$$

$$v(y+2) = x$$

$$v = \frac{x}{2+y}$$

b). $f(x, y) = 3x^2 - 2xy + y^2 - 8y$

$$f_x(x, y) = 6x - 2y$$

$$f_y(x, y) = -2x + 2y - 8$$

$$f_{xx}(x, y) = 6$$

$$f_{yy}(x, y) = 2$$

$$f_{xy}(x, y) = -2$$

$$f_x = f_y = 0 \Rightarrow 6x - 2y = 0$$

$$-2x + 2y - 8 = 0$$

$$6x = 2y$$

$$-2x + 2y = 8$$

$$3x = y$$

$$-x + y = 4$$

$$-x + 3x = 4$$

$$x(1+3)$$

$$2x = 4$$

$$x = 2 \quad y = 6$$

$$u = x$$

Critical point $(2, 6)$

$$D = f_{xx} \cdot f_{yy} - (f_{xy})^2$$

$$= 6 \cdot 2 - (-2)^2$$

$$= 12 - 4$$

$$= 8 > 0.$$

$f_{xx} = 6 > 0. \Rightarrow (2, 6)$ is a relative minima.



Module - III

$$15. a) \int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy dx.$$

$$y: x \rightarrow \infty$$

$$x: 0 \rightarrow \infty$$

$$y: 0 \rightarrow \infty$$

$$x: 0 \rightarrow y.$$

$$\Rightarrow \int_0^{\infty} \int_0^y \frac{e^{-y}}{y} dx dy.$$

$$= \int_0^{\infty} \left[\frac{e^{-y}}{y} x \right]_0^y dy.$$

$$= \int_0^{\infty} e^{-y} dy.$$

$$= -e^{-y} \Big|_0^{\infty}$$

$$= 1$$

b)

Ktunotes.in



15.6)

$$V = \iiint dx dy dz.$$

$$z = 6 - x - 2y.$$

$\therefore z$ varies from $0 \rightarrow 6 - x - 2y$.

$$z = 0 \Rightarrow x + 2y = 6.$$

$$x = 6 - 2y.$$

x varies from $0 \rightarrow 6 - 2y$.

Put $x = 0 \Rightarrow 6 = 2y$, $y = 3$.

$\therefore y$ varies from $0 \rightarrow 3$.

$$\int_0^3 \int_0^{6-2y} \int_0^{6-x-2y} dz dx dy.$$

$$= \int_0^3 \int_0^{6-2y} (6 - 2y - x) dx dy.$$

$$= \int_0^3 \left(6x - \frac{x^2}{2} - 2yx \right) \Big|_0^{6-2y} dy.$$

$$= \int_0^3 \left[6(6-2y) - \frac{(6-2y)^2}{2} - 2y(6-2y) \right] dy.$$

$$= \left[6(6y - y^2) - \frac{(6-2y)^3}{6 \times -2} - 2 \left[\frac{6y^2}{2} - \frac{2y^3}{3} \right] \right]_0^3$$

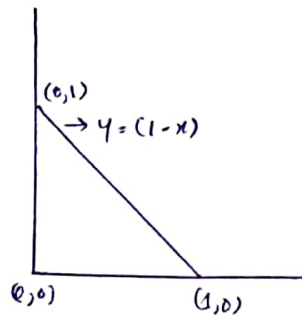
$$= \left[6(6 \times 3 - 3^2) - \frac{(6-2 \times 3)^3}{6 \times -2} - 2 \left[\frac{6 \times 3^2}{2} - \frac{2 \times 3^3}{3} \right] - \left[6(6 \times 0 - 0^2) - \frac{(6-2 \times 0)^3}{6 \times -2} - 2 \times 0 \right] \right]$$

$$= 16(18 - 9) - 2 \times 9 - 18.$$

$$= 54 - 36.$$

$$= \underline{\underline{18}}.$$

16. a)



$$y: 0 \rightarrow (1-x)$$

$$x: 0 \rightarrow 1$$

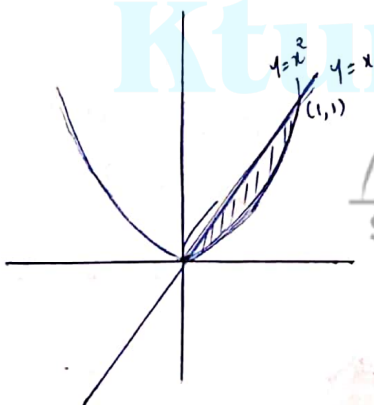
$$\int_0^1 \int_0^{1-x} xy \, dy \, dx.$$

$$= \int_0^1 \left[\frac{xy^2}{2} \right]_0^{1-x} dx.$$

$$= \int_0^1 \frac{x(1-x)^2}{2} dx.$$

$$= \frac{1}{24}$$

b)



$$y: x^2 \rightarrow x.$$

$$x: 0 \rightarrow 1$$

$$\int_0^1 \int_{x^2}^x x^2 \, dy \, dx$$

$$= \int_0^1 \left[x^2 y \right]_{x^2}^x dx$$

$$= \int_0^1 (x^3 - x^4) dx$$

$$= \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1$$

$$= \frac{1}{20}$$

17) a) i) $\sum_{k=1}^{\infty} \frac{k!}{k^k}$

$$u_k = \frac{k!}{k^k}$$

$$u_{k+1} = \frac{(k+1)!}{(k+1)^{k+1}}$$

Ratio test

$$\lim_{k \rightarrow \infty} \frac{u_{k+1}}{u_k} = \lim_{k \rightarrow \infty} \frac{\frac{(k+1)!}{(k+1)^{k+1}}}{\frac{k!}{k^k}} = \lim_{k \rightarrow \infty} \frac{(k+1)!}{(k+1)^{k+1}} \times \frac{k^k}{k!}$$

$$= \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^k = 0$$

$\therefore$ the series converges.

ii)

$$\sum_{k=1}^{\infty}$$

$$\left(\frac{k}{k+1} \right)^{k^2}$$

applying root test

$$u_k = \left(\frac{k}{k+1} \right)^{k^2}$$

$$\lim_{k \rightarrow \infty} [u_k]^{\frac{1}{k}} = \lim_{k \rightarrow \infty} \left[\left(\frac{k}{k+1} \right)^{k^2} \right]^{\frac{1}{k}} = \lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^k = 0$$

$\therefore$ the series converges.

18 a) test the convergence of $1 + \frac{1 \cdot 3}{3!} + \frac{1 \cdot 3 \cdot 5}{5!} + \dots$

23)

leaving the first term

$$a_k = \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{(2k)!}$$

$$a_{k+1} = \frac{1 \cdot 3 \cdot 5 \cdots (2k-1) (2k+1)}{(2k+2)!}$$

$$\lim_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} = \lim_{k \rightarrow \infty} \frac{1 \cdot 3 \cdot 5 \cdots (2k-1) (2k+1)}{(2k+2)!} \times \frac{(2k)!}{1 \cdot 3 \cdot 5 \cdots (2k-1)}$$

$$= \lim_{k \rightarrow \infty} \frac{(2k+1)}{(2k+2)(2k+1)} = \lim_{k \rightarrow \infty} \frac{1}{2k+2} = 0$$

$\therefore$ the series converges by ratio test

$$= \frac{1}{\pi} \left[\frac{-(-\cos n\pi)}{1+n} - \frac{(-\cos n\pi)}{1-n} + \frac{2}{1-n^2} \right]$$

$$a_n = \left[\frac{2 \cos n\pi}{1-n^2} + \frac{2}{1-n^2} \right]$$

$$= \frac{2}{\pi(1-n^2)} (\cos n\pi + 1)$$

$$= \frac{2}{\pi(1-n^2)} \quad \text{--- (2) for even } f(x) \text{ } n=2n$$

$$\therefore f(x) = \frac{4}{\pi} + \sum_{n=1}^{\infty} \frac{2 \times 2}{\pi(1-n^2)} \cos 2nx$$

$$= \frac{4}{\pi} + \sum_{n=1}^{\infty} \frac{4}{\pi(1-n^2)} \cos 2nx$$

$$f(x) = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos 2nx}{(4n^2 - 1)}$$

$$15) b) \sum_{k=1}^{\infty} (-1)^{k+1} \frac{k-2}{k(k+1)}$$

$$|u_k| = \frac{k-2}{k(k+1)} = \frac{k-2}{k^2+k}$$

$$|u_{k+1}| = \frac{(k+1)-2}{(k+1)((k+1)+1)} = \frac{k-1}{k^2+3k+2}$$

$$\lim_{k \rightarrow \infty} \frac{|u_{k+1}|}{|u_k|} = \frac{k-1}{k^2+3k+2} \times \frac{k^2+k}{k-2} = \frac{k^3+k^2-k-k}{k^3+}$$

$= \frac{1}{1} = 1 > 0$: series is not absolutely convergent
by Leibnitz test

taking $a_k = \frac{k-2}{k^2+k}$ $b_k = \frac{k}{k^2}$

limit comparison test

$$\lim_{k \rightarrow \infty} \frac{b_k}{a_k} = \frac{k-2}{k^2+k} \times \frac{k^2}{k} = \frac{k^3(1-\frac{2}{k})}{k^3-k^2} = \frac{1}{1} = 1 //$$

converges or diverges equally.

$$\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{k}{k^2} = \sum_{k=1}^{\infty} \frac{1}{k} \therefore p \text{ series with } p=1 \therefore \text{series diverges.}$$

$\therefore$ the series converges conditionally.

19) A) obtain Fourier series for the function $f(x) = |\sin x|$
 $-\pi < x < \pi$

$$f(x) = |\sin x| = \sin x$$

$$f(-x) = |\sin(-x)| = \sin x \quad \therefore f(x) \text{ is even.}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} \sin x dx \\ &= \frac{2}{\pi} [-\cos x]_0^{\pi} = \frac{2}{\pi} \times 2 = \frac{4}{\pi} \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx \\ &= \frac{2}{\pi} \int_0^{\pi} \sin x \cos nx dx \\ &= \frac{2}{\pi} \int_0^{\pi} \sin(1+n)x + \sin(1-n)x dx \end{aligned}$$

$$\therefore a_n = \frac{1}{\pi} \int_0^{\pi} \sin(1+n)x + \sin(1-n)x dx$$

$$= \frac{1}{\pi} \left[\frac{-\cos(1+n)x}{(1+n)} - \frac{\cos(1-n)x}{(1-n)} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{-\cos(1+n)\pi}{1+n} - \frac{\cos(1-n)\pi}{1-n} + \frac{1}{1-n} + \frac{1}{1+n} \right]$$

$$= \frac{1}{\pi} \left[\frac{-\cos(\pi+n\pi)}{1+n} - \frac{\cos(\pi-n\pi)}{1-n} + \frac{2}{1-n^2} \right]$$

200)

$$f(x) = x^2$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x^2 dx = \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{\pi^2}{3}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx$$

$$= \frac{2}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - 2x \left(-\frac{\cos nx}{n^2} \right) + 2 \left(-\frac{\sin nx}{n^3} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[2\pi \frac{\cos n\pi}{n^2} \right] = \frac{4(-1)^n}{n^2}$$

$$\therefore x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$$

$$= \frac{\pi^2}{3} - 4 \left[\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \frac{\cos 4x}{4^2} + \dots \right] \quad \text{--- ①}$$

Setting $x = \pi$ in ①, we get

$$\pi^2 = \frac{\pi^2}{3} - 4 \left[-\frac{1}{1^2} - \frac{1}{2^2} - \frac{1}{3^2} - \frac{1}{4^2} + \dots \right]$$

$$\Rightarrow \frac{2\pi^2}{3} = 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \right]$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \frac{\pi^2}{6}$$

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